

ML, unit-3

→ Eqⁿ of linear regression:

$$Y = mX + b$$

↓ ↓
dependent independent

↗ slope ↘ intercept

⇒ Evaluating the performance of linear regression model

• Done by MSE, MAE, and RMSE

→ These three are error metrics used to evaluate how good a linear regression model is

→ They predict how far the predicted values are from the actual values

1-) Mean absolute error (MAE)

$$\left\{ MAE = \frac{1}{n} \sum |y - \hat{y}_i| \right\}$$

Where;

y_i = actual value

$\hat{y}_i$ = predicted value

n = noⁿ of observations

MAE is the average of mean absolute errors.

→ Absolute means ignore +ve or -ve.

|error|

→ Only care about how big mistake is, not direcⁿ.

2.) Mean Squared error (MSE)

formula used :-

$$MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

• Square to punish large errors strongly.

3.) Root mean Squared Error (RMSE)

$$RMSE = \sqrt{MSE}$$

Why to use ?

→ if price is in dollars

→ MSE becomes dollar²

RMSE fixes unit issue

$$\left[MSE = \frac{RSS}{n} \right]$$

★ Regularization and regularized methods of regression

→ Regularisation is used in linear regression models to prevent overfitting

Q) Why Regression Regularisation needed?

→ In Regression, the model tries to fit the data by finding coeff.

Eg :-

$$y = b_0 + b_1 x_1 + b_2 x_2$$

Sometimes, the model learns very large coeff. like :-

$$y = 5 + 120x_1 - 3000x_2$$

This usually happens because :-

- Too many features
- Noisy data
- model tries to perfectly fit training data

→ This leads to overfitting

=> Regularisation adds a penalty to the model for having large Coefficients

- So, instead of minimizing only the error, we minimize =

$$\text{Loss} = \text{Error} + \text{Penalty}$$

- Normally, regression minimises =

$$RSS = \sum (y - \hat{y})^2$$

(Residual Sum of Squares)

with regularisation =

$$\text{Loss} = RSS + \text{penalty}$$

* Types of regularized regression

1.) Ridge regression (L2 Regularisation)

$$\text{Loss} = RSS + \lambda \sum \beta^2$$

→ β = Coefficients

λ = regularisation parameters.

=> Penalty = Square of Coeff

Effect => Reduces Coeff value but doesn't make them 0
 - feature remain in model

2.) Lasso Regression (L1 Regularisation)

$$\text{Loss} = \text{RSS} + \lambda \sum |B|$$

penalty $\rightarrow$ absolute values

Effect $\Rightarrow$ Some coeff become exactly zero

- feature with coeff 0 gets removed
- Lasso performs feature selection

3.) Elastic Net

Combination of Ridge and Lasso:

$$\text{Loss} = \text{RSS} + \lambda_1 \sum |B| + \lambda_2 \sum (B)^2$$

$\rightarrow$ used when correlated feature exists.

* Role of lambda:

$\rightarrow$ Controls how strong the penalty is:

$\rightarrow$ λ value

$$\lambda = 0$$

Small λ

largest

effect

normal L.R

small penalty

Coeff. shrink a lot

if λ is too large $\rightarrow$ underfitting

Spiral

Date _____

Unit-4, ML

* Confusion matrix

	Predicted Yes	Predicted No
Actual Yes	TP	FN
Actual No	FP	TN

$$1.) \text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$2.) \text{Precision} = \frac{TP}{TP + FP}$$

$$3.) \text{Recall} = \frac{TP}{TP + FN}$$

$$4.) \text{F1 Score} = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$5.) \text{MSE} = \frac{1}{n} \sum (y - \hat{y})^2$$

$$6.) R^2 \text{ Score} = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

where,

$$SS_{\text{res}} = \sum (y - \hat{y})^2$$

$$SS_{\text{tot}} = \sum (y - \bar{y})^2$$

$$7.) \text{ Error rate} = 1 - \text{accuracy}$$

or

$$\frac{FP + FN}{\text{Total}}$$

* ROC Curve (Receiver operating characteristic curve)

→ A ROC curve is a graph used to evaluate classification models.

→ It shows how well a model can separate +ve and -ve values.

→ Axes of ROC curve

Axis

Meaning

x-axis

False positive rate (FPR)

y-axis

True " " (TPR)

$$\rightarrow TPR = \frac{TP}{TP + FN} \quad \left. \vphantom{\frac{TP}{TP + FN}} \right\} \begin{array}{l} \text{same as} \\ \text{recall} \end{array}$$

$$\rightarrow FPR = \frac{FP}{FP + TN}$$

• measure how -ve cases were wrongly read as +ve

→ ROC curve shows how TPR changes with FPR for different thresholds

→ A better model has -

- high TPR
- low FPR

So, curve goes closer to top-left corner (closer to y axis)

→ Area under ROC curve measures model quality

↳ AUC (area under curve)

AUC value

Meaning

1

perfect model

0.9

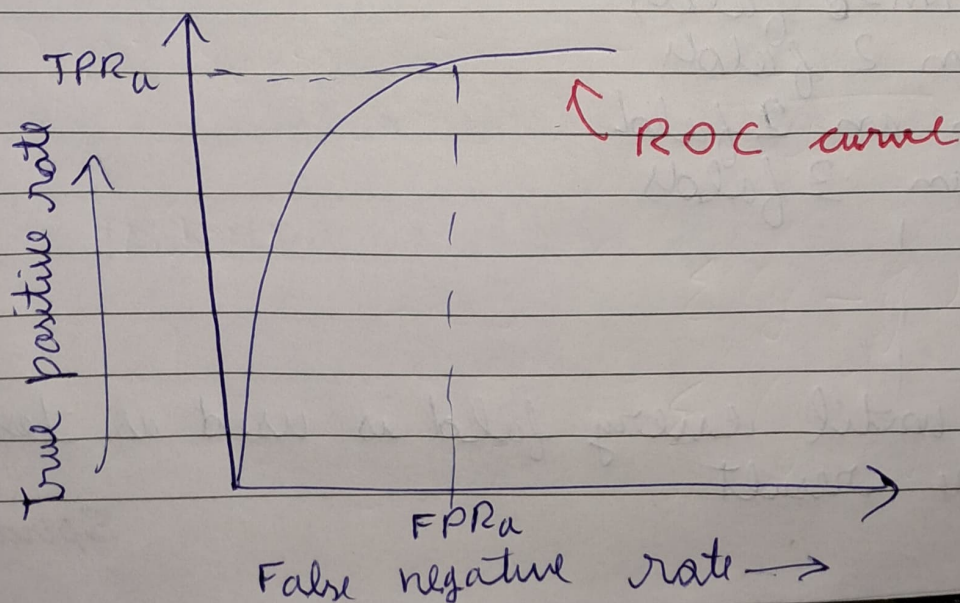
excellent

0.8

good

0.5

random guessing



* Validation methods

- Validation methods are used to test how well a machine learning model performs on unseen data
- if we only test on train data → overfitting may occur.

methods :-

1.) Train-test split

2.) K-fold cross validation

- dataset is divided into K equal parts (folds)

Eg → $K = 5$

1.) Train on 4 folds

2.) test on 1 fold

3.) Train on 3 folds

4.) test on 2 folds

5.) train on 2 folds

6.) test on 3 folds

|

|

|

- 7.) Repeat until every fold is used as test
- 8.) Avg the result

3) Leave - One - out Cross Validation (LOOCV)

- Special case of K folds where $K = n$

- If there are 100 sample, we can create 100 folds

★ Bias - Variance Decomposition

Explains why models make errors.

Total prediction error comes from three sources:

$$[\text{Error} = \text{Bias}^2 + \text{Variance} + \text{Noise}]$$

- 1) Bias: Measures how far predictions are from true values

High bias means: (1) model too simple
(2) can't capture patterns

Eg: linear model for complex data

High bias happens due to underfitting

2.) Variance : Measures how much the model changes when training data changes

High variance means [↓]

- model fits training data too closely
- model becomes too complex

Result $\rightarrow$ high variance happens due to overfitting